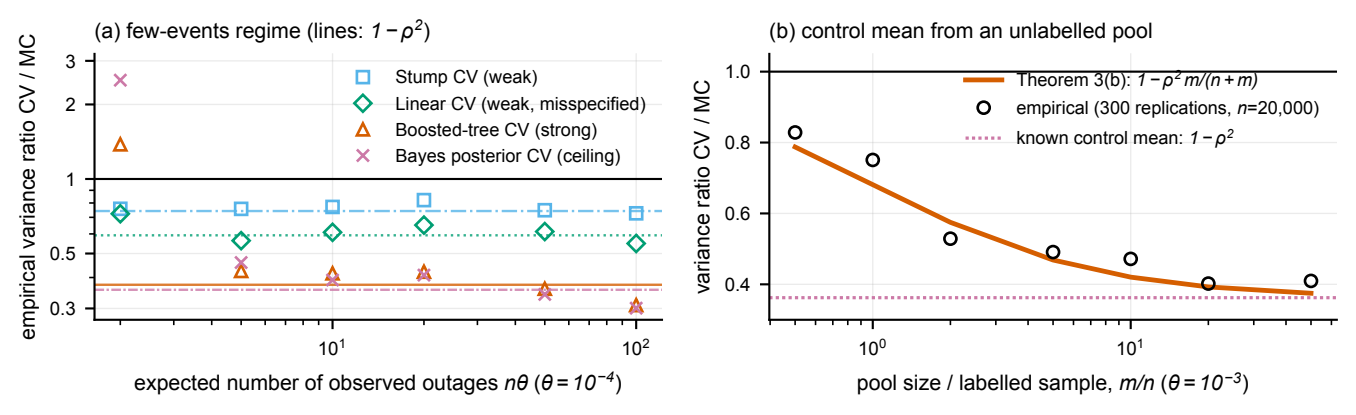
(a) few-events regime (lines: $1-\rho^2$)
empirical variance ratio CV / MC
3
2
1
0.5
0.3
Stump CV (weak)
Linear CV (weak, misspecified)
Boosted-tree CV (strong)
Bayes posterior CV (ceiling)
10^1
10^2
expected number of observed outages $n\theta$ ($\theta = 10^{-4}$)
(b) control mean from an unlabelled pool
variance ratio CV / MC
1.0
0.8
0.6
0.4
Theorem 3(b): $1-\rho^2\, m/(n+m)$
empirical (300 replications, n=20,000)
known control mean: $1-\rho^2$
10^0
10^1
pool size / labelled sample, m/n ($\theta = 10^{-3}$)